Acknowledgement

I, **Upasana Nayak**, would like to express my heartfelt gratitude to **Unified Mentor** for providing me with the opportunity to undertake this internship and work on the project titled *"Soil Farming Agent"*. This internship has been a great learning experience for me, allowing me to apply my academic knowledge in a practical environment and gain valuable industry exposure.

I am sincerely thankful to my mentors at Unified Mentor for their continuous guidance, support, and encouragement throughout the internship period. Their valuable suggestions, constructive feedback, and willingness to help at every stage played a significant role in the successful completion of this project. They not only guided me technically but also helped me improve my problem-solving approach and understanding of real-world project development.

I would also like to extend my gratitude to my faculty members and institution for providing me with the necessary knowledge, resources, and motivation to carry out this project effectively. Their constant encouragement helped me stay focused and confident while working on this project.

Furthermore, I would like to thank my peers for their support and for sharing ideas that contributed to improving my project. Discussions and interactions with them helped me gain new perspectives and enhance my work.

Finally, I am deeply thankful to my family for their unconditional support, patience, and encouragement throughout my internship journey. Their belief in me kept me motivated and helped me complete this project successfully.

Abstract

The *"Soil Farming Agent"* project is a web-based application developed to provide users with essential information about different types of soil and the crops that are most suitable for each type. Agriculture largely depends on soil quality and characteristics, and selecting the right crop for a specific soil type is crucial for achieving better productivity and sustainability. This project aims to simplify this process by offering a centralized platform where such information can be easily accessed.

The main objective of this project is to bridge the gap between soil knowledge and agricultural decision-making by presenting accurate and structured information in a user-friendly manner. The system allows users to view details about various soil types, understand their properties, and identify which crops can grow best in them

The application includes two main modules:

- **Admin Module:**
    - Responsible for managing and updating soil information
    - Adds and maintains distributor details
    - Ensures the accuracy and reliability of data with expert input
- **User Module:**
    - Allows users to register and log in
    - Enables viewing of soil details
    - Provides access to distributor information

The system is developed using modern web technologies such as:

- HTML for structuring the web pages
- CSS for styling and enhancing the user interface
- JavaScript for adding interactivity and functionality
- MongoDB for database management, authentication, and hosting

The project follows a modular and structured approach, making it easy to maintain, test, and scale in the future. Proper coding standards and practices have been followed to ensure the system is safe, reliable, and efficient.

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Introduction

Agriculture is an important part of our daily lives and plays a major role in the economy. One of the key factors that affects farming is soil. Different types of soil have different properties like texture, nutrients, and water-holding capacity. Because of this, not all crops can grow well in every type of soil. Choosing the right crop for the right soil is very important for good production.

However, many people do not have easy access to clear and organized information about soil types and suitable crops. This makes it difficult for them to make the right decisions. With the help of technology, this problem can be solved by creating a simple platform where such information is available in one place.

The *"Soil Farming Agent"* project is developed to provide a solution to this problem. It is a web-based application where users can learn about different types of soil and the crops that grow best in them. The system also provides information about crop and seed distributors, which can help users find the right resources easily.

The application is built using HTML, CSS, JavaScript, and MongoDB, making it simple to use and efficient. It also includes an admin side where all the data is managed and updated regularly.

Problem Statement

In agriculture, having the right knowledge about soil and crops is very important. But in reality, many users face difficulties in getting proper and reliable information. This creates problems in decision-making and affects overall productivity.

Some of the main problems are:

- **Lack of knowledge about soil types:**
  Many users are not aware of the different types of soil and their characteristics.
- **Wrong crop selection:**
  Due to lack of proper information, users may choose crops that are not suitable for their soil, which leads to poor results.
- **Information is not easily available:**
  Soil and crop-related information is often scattered and not presented in a simple way.
- **Difficulty in finding distributors:**
  It is not always easy to find trusted crop and seed distributors when needed.

- **No single platform:**
  There is no one system where users can find all the required information about soil, crops, and distributors together.

To solve these problems, the *"Soil Farming Agent"* project provides a simple and centralized platform where users can access all this information easily. It helps in improving awareness, making better choices, and supporting better agricultural practices.

Objectives

The main aim of the *"Soil Farming Agent"* project is to make it easier for users to understand soil types and choose the right crops. The idea is to provide all the important information in a simple and clear way so that anyone can use it without difficulty.

The key objectives of this project are:

- **To provide basic knowledge about soil:**
  - Help users learn about different types of soil
  - Make soil information easy to understand
- **To help in selecting the right crops:**
  - Guide users to choose crops based on soil type
  - Reduce mistakes in crop selection
- **To create a single platform for information:**
  - Combine soil details, crop information, and distributor data in one place
  - Save time and effort for users
- **To include distributor details:**
  - Help users find crop and seed distributors easily
  - Make resources more accessible
- **To build an easy-to-use application:**
  - Keep the design simple and user-friendly
  - Make it easy for anyone to navigate
- **To keep information updated:**
  - Allow the admin to manage and update data
  - Ensure users get correct and latest information

Scope of the Project

The *"Soil Farming Agent"* project mainly focuses on providing basic and useful information related to soil, crops, and distributors through a web application. It is designed to be simple, practical, and easy to use.

The scope of the project includes:

- **For users:**
  - Register and log in to the system
  - View different types of soil and their details
  - Get information about suitable crops
  - Check details of crop and seed distributors
- **For admin:**
  - Add and update soil information
  - Manage distributor details
  - Keep the system data accurate and organized
- **Technical scope:**
  - Web-based application
  - Built using HTML, CSS, JavaScript, and MongoDB
  - Uses MongoDB for storing data and user authentication
- **System features:**
  - Simple interface for easy understanding
  - Organized display of information
  - Basic login system for security

Limitations of the project:

- It does not provide real-time soil testing
- It does not use advanced technologies like AI for recommendations
- Information is limited to what is stored in the system
- Internet connection is required to use the application

Future improvements:

- Adding smart recommendations for crops
- Connecting with real-time data sources

- Developing a mobile version of the application
- Adding support for multiple languages

System Analysis

System analysis is the process of understanding what users need and how a system can be designed to meet those needs in a better way. In the *"Soil Farming Agent"* project, this step helps in identifying the problems users face and finding a simple and effective solution.

In real life, many users find it difficult to get clear information about soil and crops. So, this project focuses on understanding those difficulties and creating a system that makes things easier.

The main points considered during system analysis are:

- **Understanding user needs:**
    - Users want simple and clear information about soil
    - They need help in choosing the right crops
    - They also want easy access to distributor details
- **Identifying problems in the current process:**
    - Information is not properly organized
    - It is difficult to make correct decisions
    - There is no single place to find all the details
- **Deciding system features:**
    - User registration and login
    - Viewing soil and crop information
    - Accessing distributor details
    - Admin control to manage and update data
- **Making the system user-friendly:**
    - The system should be simple and easy to use
    - It should provide reliable information
    - Data should be properly managed
- **Selecting technologies:**

    - HTML, CSS, and JavaScript for building the interface
    - MongoDB for storing data and handling authentication

This analysis helps in building a system that is simple, useful, and easy to manage for both users and admin.

Existing System

In the current scenario, there is no proper system where users can find all the information about soil, crops, and distributors in one place. Most of the time, people depend on traditional methods or different sources to get this information.

The existing system works in the following way:

- **Dependence on personal knowledge:**
    - Users rely on their own experience or advice from others
    - Information may not always be correct or updated
- **Information is scattered:**
    - Details are available in books, websites, or from experts
    - There is no single platform that combines everything
- **Difficult to access information:**
    - Not everyone can easily reach experts
    - Information may be hard to understand for some users
- **No proper data management:**
    - Information is not stored in an organized digital form
    - It takes time to find the required details
- **Lack of distributor information:**
    - Users may find it hard to locate trusted distributors
    - No structured system to provide such details

Problems with the existing system:

- It takes more time and effort to collect information
- There is a higher chance of making wrong decisions
- The process is not very efficient

- Information is not easily accessible or updated

Because of these issues, there is a need for a better system. The *"Soil Farming Agent"* project is created to solve these problems by providing all the necessary information in a simple and organized way.

Proposed System

The *"Soil Farming Agent"* project is created to solve the problems faced in the existing system by providing a simple and organized solution. It is a web-based application where users can easily find information about soil types, suitable crops, and distributor details—all in one place.

The main idea behind this system is to make everything आसान and easily accessible, so users don't have to depend on multiple sources for information.

The proposed system includes the following features:

- **All-in-one platform:**
  - Users can find soil, crop, and distributor information in one place
  - Saves time and effort
- **Easy to use interface:**
  - Simple design that anyone can understand
  - Smooth navigation for better user experience
- **User side features:**
  - Users can register and log in
  - View soil details and suitable crops
  - Check distributor information easily
- **Admin side features:**
  - Admin can add and update soil details
  - Admin can manage distributor information
  - Keeps the data accurate and updated
- **Proper data management:**
  - All data is stored using MongoDB
  - Easy to update and access information anytime

- **Better decision-making:**
    - Helps users choose the right crops
    - Provides reliable and useful information

System Requirements

System requirements explain what is needed to build and run the project properly. These are divided into two types: functional requirements and non-functional requirements.

Functional Requirements

These are the main features that the system must provide:

- **User registration and login:**
    - Users should be able to create an account
    - Secure login system to access the application
- **View soil details:**
    - Users can see different types of soil
    - Basic information about each soil type is displayed
- **View crop information:**
    - Users can check which crops are suitable for specific soil
- **View distributor details:**
    - Users can access information about crop and seed distributors
- **Admin control:**
    - Admin can log in to the system
    - Admin can add, update, or delete data
- **Data storage:**
    - All information is stored in the database
    - Data can be easily managed and retrieved

Non-Functional Requirements

These describe how the system should perform:

- **Usability:**

    - The system should be simple and easy to use
    - Users should not face difficulty while using it
- **Performance:**
    - The application should load quickly
    - Information should be displayed without delay
- **Security:**
    - User data should be safe
    - Proper login and authentication should be maintained
- **Reliability:**
    - The system should work smoothly without errors
    - Data should remain correct and consistent
- **Maintainability:**
    - The code should be easy to update
    - Changes should be simple to implement in the future
- **Scalability:**
    - The system should handle more data if needed later
- **Portability:**
    - The application should work on different devices and browsers

These requirements make sure that the system is not only working properly but is also easy to use, secure, and ready for future improvements.

System Design

System design explains how the *"Soil Farming Agent"* project is planned and how all its parts work together. The main goal of the design is to keep the system simple, clear, and easy to use for both users and admin.

The project is designed in a way where different parts of the system are separated, so it becomes easier to build, understand, and maintain. This also helps in making future changes without much difficulty.

The main parts of the system design are:

- **Frontend (User Interface):**

    - Built using HTML, CSS, and JavaScript
    - Provides pages like login, registration, and information display
    - Designed to be simple and user-friendly
- **Functionality (JavaScript):**
    - Handles user actions like login, viewing data, etc.
    - Connects the frontend with the database
    - Controls how data is shown and managed
- **Database (MongoDB):**
    - Stores all the data like user details, soil information, and distributor details
    - Makes it easy to save and retrieve data
- **Authentication:**
    - Managed using MongoDB
    - Provides secure login and registration for users and admin
- **Admin Module:**
    - Admin can add and update soil and distributor details
    - Keeps the system data accurate
- **User Module:**
    - Users can view soil and crop information
    - Can also check distributor details

Important design points:

- Simple and clean layout
- Easy navigation
- Secure login system
- Easy to manage and update

Architecture Diagram

The architecture of the *"Soil Farming Agent"* project shows how different parts of the system are connected and how data flows between them. It follows a simple structure where the frontend directly connects to MongoDB.

Main components:

- **User/Admin (Client side):**
  - Access the system through a web browser
  - Interact with the application
- **Frontend:**
  - Built using HTML, CSS, and JavaScript
  - Takes input from users and displays data
- **MongoDB (Backend services):**
  - Handles authentication (login and signup)
  - Stores and manages all data
- **Database:**
  - Contains soil details
  - Contains distributor information
  - Contains user data

How the system works:

1. User or admin opens the website
2. Logs in or registers into the system
3. The frontend sends a request to MongoDB
4. MongoDB processes the request
5. Data is sent back and shown on the screen
6. Admin can update data, which is saved in the database

Key points of the architecture:

- Simple and easy to understand
- No complex backend server required
- Fast data access
- Secure user authentication

Database Design

The database design of the *"Soil Farming Agent"* project is kept simple and organized so that all the information can be stored and accessed easily. Since MongoDB is used, the data is stored in a structured format that makes it easy to manage and update.

The database mainly stores three types of information:

- **User Data:**
  - Contains details of registered users
  - Includes basic information like name and email
  - Login details are handled securely through MongoDB authentication
- **Soil Data:**
  - Stores different types of soil
  - Includes soil name, its characteristics, and suitable crops
- **Distributor Data:**
  - Contains information about crop and seed distributors
  - Includes details like name, contact information, and location

Basic structure of the database:

- **Users:**
  - user_id
  - name
  - email
- **Soil:**
  - soil_id
  - soil_name
  - characteristics
  - suitable_crops
- **Distributors:**
  - distributor_id
  - name
  - contact_details
  - location

Important points:

- Data is stored in an organized way
- Easy to add, update, or delete information
- Fast access to data using MongoDB
- Secure handling of user information
- Can be expanded easily in the future

Implementation

Implementation is the stage where the project is actually developed and turned into a working application. In the *"Soil Farming Agent"* project, all the planned features are built using simple web technologies.

The project is developed using:

- **HTML:**
  - Used to create the basic structure of the web pages
- **CSS:**
  - Used to design the pages and make them look clean and user-friendly
- **JavaScript:**
  - Adds functionality to the system
  - Handles user actions like login, data viewing, etc.
- **MongoDB:**
  - Used for storing data
  - Handles user login and authentication
  - Helps in managing the backend easily

Steps followed during development:

- Creating the layout of the application using HTML and CSS
- Adding functionality using JavaScript
- Connecting the application with MongoDB
- Implementing login and registration system

- Developing admin features to manage data
- Testing the application to fix errors

Features implemented:

- User registration and login
- Viewing soil details and suitable crops
- Viewing distributor information
- Admin panel for adding and updating data

Key points:

- Code is written in a simple and modular way
- Easy to understand and maintain
- Focus on user-friendly design
- Proper use of MongoDB for backend support

Overall, the implementation part converts the idea into a real application that users can easily access and use.

Technologies Used

The *"Soil Farming Agent"* project is built using simple and commonly used web technologies. These technologies make the application easy to develop, use, and manage.

The main technologies used are:

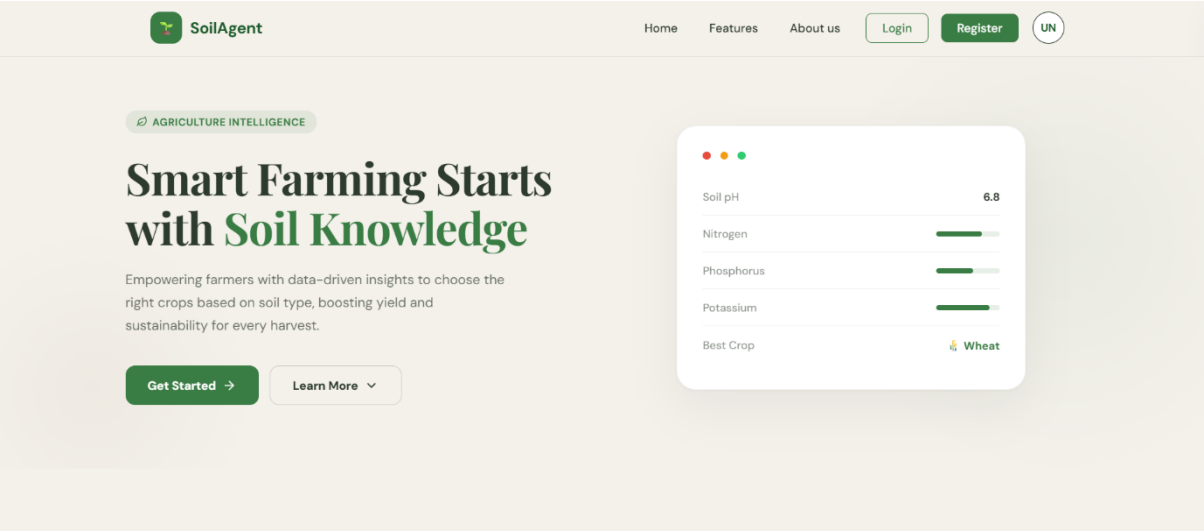
- **HTML (HyperText Markup Language):**
    - Used to create the basic structure of the website
    - Helps in arranging content like text, forms, and buttons
- **CSS (Cascading Style Sheets):**
    - Used to design the web pages
    - Makes the application look neat and user-friendly
- **JavaScript:**
    - Adds functionality to the system
    - Handles user actions like login, displaying data, and interactions

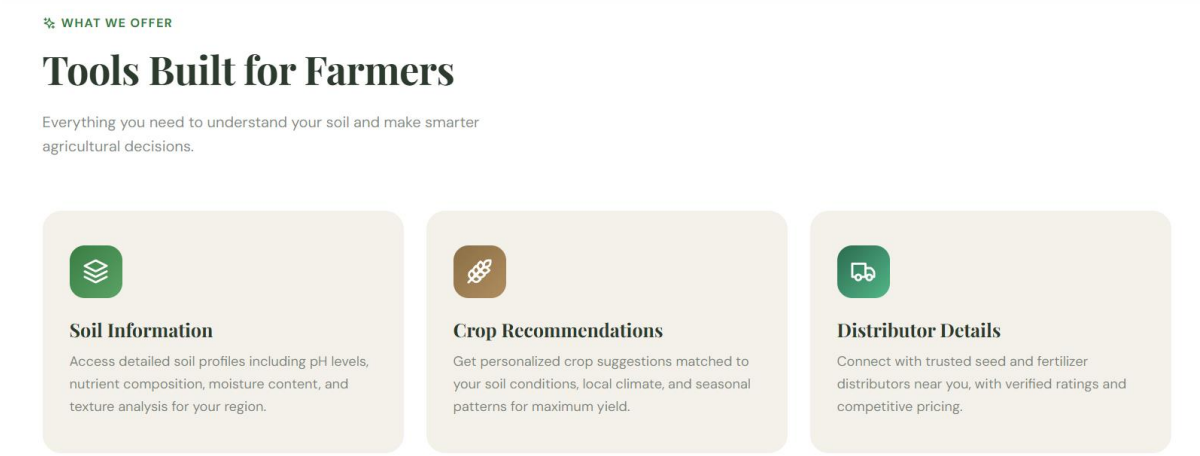
- **MongoDB:**
    - Used as the backend of the project
    - Stores all data like user details, soil information, and distributor data
    - Handles login and authentication
    - Makes data management simple
- **GitHub:**
    - Used to store and manage the project code
    - Helps in tracking changes and maintaining the project

Key points:

- Technologies are easy to use and understand
- Suitable for building web applications
- Help in smooth performance of the system
- Allow future improvements and updates

Screenshots / Output





Testing & Results

Testing is done to make sure that the project works properly and all features are functioning as expected. In the *"Soil Farming Agent"* project, different parts of the system were tested step by step.

Testing done in the project:

- **Login and Registration:**
  - Checked if users can sign up and log in successfully
  - Verified that authentication works properly
- **Soil Information:**
  - Checked if soil details are displayed correctly
  - Ensured data is properly fetched from the database
- **Distributor Information:**
  - Verified that distributor details are shown correctly
  - Checked if data is accurate
- **Admin Features:**
  - Tested adding and updating data
  - Ensured changes are saved properly
- **User Interface:**
  - Checked if all pages load correctly
  - Made sure the system is easy to use
- **Error Handling:**
  - Tested invalid inputs

    - Ensured proper messages are shown to users

Results:

- The system works properly without major issues
- Users can easily register, log in, and view information
- Data is stored and displayed correctly
- Admin can manage data without problems
- The interface is simple and easy to understand

Overall outcome:

- The project meets its main objectives
- It provides useful information in a simple way
- Helps users make better decisions
- Shows how technology can support agriculture

The testing shows that the system is working well and can be improved further in the future if needed.

Optimization Techniques

In the *"Soil Farming Agent"* project, some simple optimization techniques were used to make the system work smoothly and efficiently. The main aim was to keep the application fast, easy to use, and not too heavy.

The following methods were used:

- **Efficient data usage:**
    - Only necessary data is fetched from MongoDB
    - Helps in reducing loading time
- **Simple code structure:**
    - Code is written in a clean and modular way
    - Makes it easier to understand and update

- **Lightweight design:**
    - Avoided heavy graphics and complex layouts
    - Helps the application run smoothly on all devices
- **Use of MongoDB:**
    - Provides fast data storage and retrieval
    - Reduces the need for a complex backend
- **Input validation:**
    - User inputs are checked before processing
    - Helps in avoiding errors and improving performance
- **Smooth user experience:**
    - Simple navigation between pages
    - Quick response while using the application

Overall:

- Improves speed and performance
- Makes the system more reliable
- Helps in easy maintenance and future improvements

Challenges Faced

While working on the *"Soil Farming Agent"* project, a few challenges were faced during development. These challenges helped in learning new things and improving practical skills.

Some of the main challenges are:

- **Learning MongoDB:**
    - It was a bit difficult in the beginning to understand how MongoDB works
    - Took time to learn database and authentication features
- **Connecting frontend and backend:**
    - Linking JavaScript with MongoDB was not easy at first
    - Needed proper understanding of data flow
- **Managing data properly:**

    - Organizing soil and distributor data correctly
    - Making sure data is stored and displayed properly
- **Login system:**
    - Implementing user authentication
    - Handling errors like wrong login details
- **Designing the interface:**
    - Making the application simple but effective
    - Ensuring users can easily use it
- **Fixing errors:**
    - Debugging code during development
    - Solving unexpected issues
- **Managing time:**
    - Balancing learning and completing the project
    - Finishing everything within the deadline

What I learned:

- Better understanding of web development
- Practical knowledge of MongoDB
- Improved problem-solving skills
- Experience of working on a real project

These challenges made the learning process more meaningful and helped in building a better project.

Future Scope

The *"Soil Farming Agent"* project is a basic system that works well, but it can be improved a lot in the future. With more features and better technology, it can become more useful and practical for users.

Some improvements that can be added in the future are:

- **Smart crop suggestions:**
    - Use advanced technologies like AI to suggest better crops

    - Give more accurate results based on user needs
- **Real-time soil information:**
    - Connect with tools or data sources to get live soil updates
    - Help users make better decisions
- **Mobile application:**
    - Create a mobile version of the app
    - Make it easier to use anytime and anywhere
- **Multiple language support:**
    - Add different languages
    - Make it easier for more people to use the system
- **Weather updates:**
    - Add weather information
    - Help users plan crops based on climate
- **Better design:**
    - Improve the look and feel of the application
    - Make it more interactive
- **More data:**
    - Add more soil types and crop details
    - Include more distributor information
- **Location-based features:**
    - Show nearby distributors based on user location
    - Make the system more helpful in real situations

These improvements can make the project more advanced and useful in the future.

Conclusion

The *"Soil Farming Agent"* project was created to make it easier for users to understand soil types and choose the right crops. The main aim was to provide all the important information in a simple and organized way.

In this project, a web application was developed using HTML, CSS, JavaScript, and MongoDB. It allows users to view soil details, check suitable crops, and find distributor information. The admin can also manage and update the data easily.

This project helps in reducing the difficulty of finding useful information and supports better decision-making. It also shows how technology can be used in agriculture to solve real-life problems.

While working on this project, many new things were learned, such as using MongoDB, managing data, and building a complete application. It also helped in improving problem-solving skills and understanding how real projects are developed.

Overall, the project meets its goals and works successfully. It also has a lot of scope for improvement in the future, which can make it even more useful.

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